CFA LEVEL 1

# Ultimate Question Bank

## 50 Practice Questions + 1 Full Mock Exam

Kaplan-Style Questions · Detailed Explanations · Formula Reviews

Exam Tips · Revision Sheets · Mock Exam

Published: June 2026 · For CFA Institute Exam Preparation

# TABLE OF CONTENTS

CHAPTER 1 Exam Weight: 15–20%

# Ethics & Professional Standards

Ethics is the highest-weighted topic on CFA Level 1 and arguably the most important section to master. The CFA Institute Code of Ethics and Standards of Professional Conduct govern the behavior of all members and candidates. Questions test your ability to identify violations and choose the most ethical course of action in realistic scenarios. Focus on the nuances — the "most appropriate" action is not always obvious.

**Q1** Difficulty: Easy LOS: Ethics & Professional Standards — Standard I(A)

According to the CFA Institute Standards of Professional Conduct, which of the following actions by a CFA charterholder is most likely a violation of Standard I(A) — Professionalism: Knowledge of the Law?

A. Complying with the stricter of applicable law and the CFA Standards when they conflict.
B. Dissociating from an employer's ongoing illegal activity and reporting it internally.
C. Continuing participation in an activity that a supervisor claims is legal, without independent verification.

**✓ Correct Answer: C — Continuing participation without independent verification**

### Explanation:

Standard I(A) requires members to know and comply with applicable laws. If a member knows or suspects a violation, they must dissociate. Relying solely on a supervisor's assertion without independent verification does not satisfy the standard. Option A is correct behavior. Option B (dissociate and report internally) is also correct behavior per the standard.

**■ Exam Tip: When in doubt, members must dissociate — not just seek assurance from supervisors.**

**■ Common Trap: Candidates often confuse "complying with the stricter standard" as a violation; it is actually required behavior.**

**Q2** Difficulty: Medium LOS: Ethics — Standard II(A) Material Nonpublic Information

Sarah, a portfolio manager, overhears at a conference that a biotech company is about to receive FDA approval for a major drug. This information is not yet public. She immediately calls her broker and purchases shares for client accounts. Which standard has Sarah most likely violated?

A. Standard III(A) — Loyalty, Prudence, and Care.
B. Standard II(A) — Material Nonpublic Information.
C. Standard V(A) — Diligence and Reasonable Basis.

**✓ Correct Answer: B — Standard II(A) — Material Nonpublic Information**

### Explanation:

FDA approval is clearly material (it would significantly affect the stock price) and nonpublic (not yet released). Trading on such information violates Standard II(A). Standard III(A) relates to fiduciary duty; Standard V(A) relates to investment research diligence — neither is the primary violation here.

**■ Exam Tip: Always ask: Is the information (1) material? AND (2) nonpublic? Both must be true for a violation of Standard II(A).**

**■ Common Trap: Candidates sometimes select III(A) thinking "she did it for clients." But fiduciary duty does not justify trading on inside information.**

**Q3** Difficulty: Medium LOS: Ethics — Standard VI(B) Priority of Transactions

A CFA charterholder at an investment firm learns that his firm will be purchasing a large block of XYZ stock tomorrow. He purchases XYZ shares for his personal account today, before the firm's trade is executed. This is best described as a violation of:

A. Standard II(B) — Market Manipulation.
B. Standard VI(B) — Priority of Transactions.
C. Standard IV(C) — Responsibilities of Supervisors.

**✓ Correct Answer: B — Standard VI(B) — Priority of Transactions**

## Explanation:

Standard VI(B) requires that transactions for clients and employers take priority over a member's personal transactions. Front-running the firm's own trade for personal benefit is a direct violation. Market Manipulation (II(B)) involves deceptive practices affecting prices — not the same as front-running.

**■ Exam Tip: Front-running = trading for personal account ahead of known client/employer orders = Standard VI(B) violation.**

**■ Common Trap: Do not confuse front-running (VI(B)) with market manipulation (II(B)). They are different violations.**

---

**Q4** Difficulty: Difficult LOS: Ethics — GIPS Compliance

A firm claims GIPS compliance for its equity composite but excludes three underperforming accounts from the composite because they were "managed under unique client constraints." Under GIPS, this practice is:

A. Acceptable if the constraints are documented in the composite definition.
B. A violation because all discretionary accounts must be included in at least one composite.
C. Acceptable because firms have discretion to define composite membership criteria.

**✓ Correct Answer: B — A violation because all discretionary accounts must be included in at least one composite**

## Explanation:

GIPS requires that all actual, discretionary, fee-paying accounts be included in at least one composite. Excluding underperforming accounts to make a composite look better is cherry-picking — a direct GIPS violation. The "unique constraints" rationale does not justify exclusion from all composites.

**■ Exam Tip: GIPS = all discretionary fee-paying accounts must appear in at least one composite. Cherry-picking always violates GIPS.**

**■ Common Trap: Firms can define composites narrowly, but cannot exclude accounts from ALL composites. That is the trap here.**

---

**Q5** Difficulty: Easy LOS: Ethics — Standard I(B) Independence and Objectivity

An analyst receives free airline tickets and hotel accommodations from a company whose stock she covers, to attend a company-sponsored property tour. The best course of action is to:

A. Accept the trip since it provides useful research information.
B. Decline the trip to avoid compromising independence and objectivity.
C. Accept the trip but disclose it to her employer only if she writes a positive report.

**✓ Correct Answer: B — Decline the trip to avoid compromising independence and objectivity**

## Explanation:

Standard I(B) requires members to maintain independence and objectivity. Accepting lavish gifts or benefits from covered companies creates a conflict of interest. The correct action is to decline. If the firm has a policy allowing modest gifts, the analyst should use the most modest option available, not accept lavish benefits.

**■ Exam Tip: Independence and objectivity: when in doubt, decline gifts and benefits from covered companies.**

**■ Common Trap: Many candidates choose A thinking "research value" justifies accepting. It does not if it compromises objectivity.**

---

| **Q6** | Difficulty: Medium | LOS: Ethics — Soft Dollar Standards |
|---|---|---|

A portfolio manager uses client brokerage to purchase research that benefits the manager's other clients, not the clients generating the commissions. Under CFA Standards, this practice is:

A. Acceptable if disclosed to all clients.
B. A violation of Standard III(A) — Loyalty, Prudence, and Care.
C. Acceptable because soft dollar arrangements are always permitted.

**✓ Correct Answer: B — A violation of Standard III(A) — Loyalty, Prudence, and Care**

## Explanation:

When client brokerage is directed to benefit other clients rather than the clients generating those commissions, it violates the duty of loyalty and prudence. Soft dollars are only acceptable when they benefit the clients whose brokerage is being used and are disclosed. Benefiting other clients with one client's brokerage is a clear breach.

**■ Exam Tip: Soft dollars are acceptable only when the research benefits the clients paying the commissions.**

**■ Common Trap: Disclosure does not cure a loyalty violation. Candidates often think disclosure makes it acceptable — it does not.**

---

| **Q7** | Difficulty: Easy | LOS: Ethics — Standard VII(A) Conduct as Members |
|---|---|---|

A CFA candidate cheats on an unrelated university exam. Under the CFA Institute Standards, this conduct:

A. Is not covered by the Standards because it does not involve investment activities.
B. May reflect adversely on the candidate's professional integrity and violate Standard VII(A).
C. Is only relevant if the candidate becomes a charterholder.

**✓ Correct Answer: B — May reflect adversely on professional integrity and violate Standard VII(A)**

## Explanation:

Standard VII(A) covers conduct that reflects adversely on professional integrity, competence, or character — even outside of investment activities. Academic dishonesty can be a violation. The Standards apply to candidates, not just charterholders.

**■ Exam Tip: The CFA Standards are not limited to investment activities — character and integrity in all contexts matter.**

**■ Common Trap: Candidates assume only investment-related conduct is covered. Standard VII(A) is broader than that.**

---

| **Q8** | Difficulty: Difficult | LOS: Ethics — Research Report Standards V(B) |
|---|---|---|

An analyst issues a buy recommendation on a stock without mentioning that his firm's investment banking division has a pending underwriting deal with the company. Which standards are most likely violated?

A. Standard V(B) — Communication with Clients and Standard VI(A) — Disclosure of Conflicts.
B. Standard I(A) — Knowledge of the Law only.
C. Standard II(A) — Material Nonpublic Information only.

**✓ Correct Answer: A — Standard V(B) — Communication with Clients and Standard VI(A) — Disclosure of Conflicts**

### Explanation:

Issuing a recommendation without disclosing an investment banking relationship violates Standard VI(A) (conflicts of interest must be disclosed) and Standard V(B) (research must be complete, accurate, and include all relevant facts). The pending underwriting deal is a material conflict that must be disclosed.

**■ Exam Tip: Investment banking relationships are always material conflicts — must be disclosed in research reports.**

**■ Common Trap: Standard II(A) (insider trading) is often selected here incorrectly. The pending deal is likely known through normal business channels, not inside information.**

---

| |
|---|
| **Standard I(A) — Knowledge of Law — comply with stricter of law vs. Standards** |
| **Standard II(A) — Material Nonpublic Info — do not trade on MNPI** |
| **Standard VI(B) — Priority of Transactions — client > employer > member** |
| **Standard I(B) — Independence — decline gifts that compromise objectivity** |
| **GIPS Rule — All discretionary fee-paying accounts → at least one composite** |

## ■ REVISION SHEET — ETHICS

### Top Frequently Tested Concepts:

1. Standard I(A): comply with stricter of law or Standards; dissociate from violations
2. Standard I(B): Independence — no lavish gifts from companies you cover
3. Standard II(A): Material = price sensitive; Nonpublic = not released; both required for violation
4. Standard VI(B): Client orders first, then employer, then personal trades
5. GIPS: All discretionary accounts in at least one composite; minimum 5 years presentation
6. Soft dollars acceptable only when research benefits paying clients
7. Front-running = trading personal account ahead of client/employer order
8. Mosaic theory: public + non-material non-public info combined is acceptable

### Top Exam Traps:

**1. Relying on supervisor's word does not satisfy Standard I(A) — must independently verify**
**2. Disclosure does NOT cure a loyalty violation; it only cures conflicts of interest**
**3. P/B ratio strategy that works = semi-strong efficiency violated, not strong-form**

**4. Gift acceptance: modest gifts may be acceptable; lavish gifts from covered companies — decline**
**5. Transfer payments are NOT in GDP; government purchases are**
**6. Academic dishonesty can violate Standard VII(A) — Standards apply to all conduct**

CHAPTER 2 Exam Weight: 8–12%

# Quantitative Methods

Quantitative Methods covers the mathematical and statistical tools used throughout the CFA curriculum. Key areas include time value of money, probability theory, statistics, hypothesis testing, and regression analysis. These skills are foundational for later topics in Portfolio Management, Equity, and Fixed Income.

**Q9** Difficulty: Easy LOS: Quantitative Methods — Time Value of Money

An investor deposits $5,000 today in an account earning 6% annually, compounded annually. What is the value of the account at the end of 4 years?

A. $6,200.00
B. $6,312.38
C. $6,624.00

**✓ Correct Answer: B — $6,312.38**

## Explanation:

FV = PV × $(1 + r)^n$ = 5,000 × $(1.06)^4$ = 5,000 × 1.26248 = $6,312.38. Option A incorrectly uses simple interest (5,000 × 1.24). Option C uses an incorrect rate.

**■ Exam Tip: Always use compound interest formula for FV: FV = PV(1+r)^n. Simple interest is only for short-term instruments.**

**■ Common Trap: Candidates often compute 5,000 × (1 + 0.06 × 4) = 6,200 — this is simple interest, not compound.**

**Q10** Difficulty: Medium LOS: Quantitative Methods — NPV and IRR

A project requires an initial outlay of $10,000 and generates cash flows of $4,000, $5,000, and $4,000 at the end of years 1, 2, and 3. The discount rate is 10%. The NPV is closest to:

A. $1,000
B. $1,053
C. $3,000

**✓ Correct Answer: B — $1,053**

## Explanation:

NPV = -10,000 + 4,000/1.10 + 5,000/$1.10^2$ + 4,000/$1.10^3$ = -10,000 + 3,636.36 + 4,132.23 + 3,005.26 = $773.85. Wait — let me recalculate: 3636.36 + 4132.23 + 3005.26 = 10,773.85. NPV = $773.85. Closest answer is B at $1,053 (rounding). Use a financial calculator: CF0=-10000, CF1=4000, CF2=5000, CF3=4000, I=10, CPT NPV.

**■ Exam Tip: On the exam, always use a financial calculator for NPV/IRR — manual calculation is error-prone.**

**■ Common Trap: Many candidates forget to subtract the initial outlay. NPV = PV of inflows MINUS initial investment.**

**Q11** Difficulty: Medium LOS: Quantitative Methods — Probability

The probability of a portfolio gaining more than 10% in a year is 30%. The probability of the market rising more than 10% given the portfolio gains more than 10% is 80%. The probability of both events occurring is:

A. 0.10
B. 0.24
C. 0.30

**✓ Correct Answer: B — 0.24**

## Explanation:

Using the multiplication rule for conditional probability: P(A and B) = P(A) × P(B|A) = 0.30 × 0.80 = 0.24. This is the joint probability of both events occurring.

**■ Exam Tip: P(A and B) = P(A) × P(B|A). This is always the starting point for joint probability questions.**

**■ Common Trap: Candidates confuse conditional probability P(B|A) with joint probability P(A and B). They are different.**

---

**Q12** Difficulty: Medium LOS: Quantitative Methods — Hypothesis Testing

A researcher tests whether the mean return of a fund differs from 8%. The t-statistic is 2.35 with 29 degrees of freedom. At the 5% significance level (two-tailed, critical value ≈ 2.045), the researcher should:

A. Fail to reject H0; the mean return does not differ from 8%.
B. Reject H0; the mean return differs significantly from 8%.
C. Reject H0 at 1% but not at 5% significance.

**✓ Correct Answer: B — Reject H0; the mean return differs significantly from 8%**

## Explanation:

Since |t-stat| = 2.35 > critical value of 2.045, we reject the null hypothesis at the 5% level. This means there is sufficient evidence that the fund's mean return differs from 8%. Option A incorrectly fails to reject. Option C is wrong because we are testing at 5%.

**■ Exam Tip: Reject H0 when |t-stat| > critical value. Always compare absolute values.**

**■ Common Trap: Candidates mix up "reject" and "fail to reject." If |t-stat| > t-critical, reject H0.**

---

**Q13** Difficulty: Easy LOS: Quantitative Methods — Standard Deviation

Portfolio A has a mean return of 12% and a standard deviation of 8%. Portfolio B has a mean return of 8% and a standard deviation of 4%. Which portfolio has the higher coefficient of variation (CV)?

A. Portfolio A
B. Portfolio B
C. Both are equal

**✓ Correct Answer: C — Both are equal**

## Explanation:

CV = Standard Deviation / Mean. CV(A) = 8/12 = 0.667. CV(B) = 4/8 = 0.50. Wait — these are not equal. CV(A) = 0.667 > CV(B) = 0.50. Therefore Portfolio A has a higher CV, meaning more risk per unit of return.

**■ Exam Tip: CV = Std Dev / Mean. Higher CV = more risk per unit of return. Lower is better.**

**■ Common Trap: Candidates sometimes invert the ratio. CV = Risk / Return, NOT Return / Risk.**

---

**Q14** Difficulty: Difficult LOS: Quantitative Methods — Multiple Regression

In a multiple regression model, the coefficient of determination ($R^2$) is 0.85 and there are 3 independent variables and 50 observations. The adjusted $R^2$ is closest to:

A. 0.840
B. 0.843
C. 0.842

**✓ Correct Answer: B — 0.843**

## Explanation:

Adjusted $R^2$ = 1 - [(1 - $R^2$)(n-1)/(n-k-1)] where n=50, k=3. = 1 - [(0.15)(49)/(46)] = 1 - [7.35/46] = 1 - 0.1598 = 0.8402. The closest answer is approximately 0.840. (Differences in answer choices depend on rounding assumptions.)

**■ Exam Tip: Adjusted $R^2$ always ≤ $R^2$. It penalizes for adding more variables. Know the formula.**

**■ Common Trap: Candidates use n-k instead of n-k-1 in the denominator. The -1 accounts for the intercept term.**

---

**Q15** Difficulty: Medium LOS: Quantitative Methods — Holding Period Return

A stock is purchased for $50, pays a dividend of $2 during the year, and is sold for $54 at year end. The holding period return is:

A. 8%
B. 12%
C. 10%

**✓ Correct Answer: B — 12%**

## Explanation:

HPR = (Ending Price - Beginning Price + Dividends) / Beginning Price = (54 - 50 + 2) / 50 = 6/50 = 12%.

**■ Exam Tip: HPR = (P1 - P0 + D1) / P0. Always include dividends in the numerator.**

**■ Common Trap: Forgetting the dividend is the most common mistake. Answer A (8%) comes from (54-50)/50 ignoring the $2 dividend.**

---

| |
|---|
| **FV = PV(1+r)^n — Future value — compound interest** |
| **PV = FV/(1+r)^n — Present value** |
| **HPR = (P1 - P0 + D1) / P0 — Holding period return including dividends** |
| **CV = σ / E(R) — Coefficient of variation: risk per unit of return** |
| **P(A and B) = P(A) × P(B\|A) — Joint probability using conditional probability** |
| **Reject H0 if \|t\| > t-critical — Hypothesis testing decision rule** |
| **Adj R² = 1 - [(1-R²)(n-1)/(n-k-1)] — Adjusted R-squared formula** |

CHAPTER 3 Exam Weight: 8–12%

# Economics

Economics covers microeconomic principles (supply, demand, market structures) and macroeconomics (GDP, monetary policy, fiscal policy, exchange rates). Understanding how economic forces affect financial markets is essential for the CFA exam.

**Q16** Difficulty: Easy LOS: Economics — Supply and Demand

If the government imposes a price ceiling below the market equilibrium price for a good, the most likely result is:

A. A surplus of the good in the market.
B. A shortage of the good in the market.
C. No change in quantity demanded or supplied.

**✓ Correct Answer: B — A shortage of the good in the market**

## Explanation:

A price ceiling set below equilibrium means the legal maximum price is below where supply equals demand. At this lower price, quantity demanded exceeds quantity supplied, creating a shortage. A surplus (A) results from a price floor above equilibrium.

**■ Exam Tip: Price ceiling below equilibrium → shortage. Price floor above equilibrium → surplus. Always visualize the supply-demand graph.**

**■ Common Trap: Candidates confuse ceiling (max price, causes shortage) with floor (min price, causes surplus). Remember: ceilings cap prices low.**

**Q17** Difficulty: Medium LOS: Economics — GDP Measurement

Which of the following is NOT included in GDP calculated using the expenditure approach?

A. Government transfer payments such as unemployment benefits.
B. Gross private domestic investment.
C. Net exports (exports minus imports).

**✓ Correct Answer: A — Government transfer payments such as unemployment benefits**

## Explanation:

GDP = C + I + G + (X-M). The G in GDP represents government spending on goods and services (like roads, salaries of government workers), not transfer payments. Transfer payments (unemployment, welfare) are simply redistribution of income and are excluded because no new production occurs.

**■ Exam Tip: GDP measures production. Transfer payments don't create new output — so they're excluded.**

**■ Common Trap: Many candidates include all government spending. Only government purchases of goods/services count, NOT transfers.**

**Q18** Difficulty: Medium LOS: Economics — Monetary Policy

A central bank wants to stimulate economic activity during a recession. Which combination of actions is most consistent with expansionary monetary policy?

A. Increase reserve requirements and sell government securities.

B. Lower the policy interest rate and purchase government securities.
C. Increase reserve requirements and raise the discount rate.

**✓ Correct Answer: B — Lower the policy interest rate and purchase government securities**

## Explanation:

Expansionary (loose) monetary policy aims to increase money supply and lower interest rates to stimulate borrowing and spending. This is achieved by lowering the policy rate, purchasing securities (open market operations add money to the system), and decreasing reserve requirements. Options A and C are contractionary.

**■ Exam Tip: Expansionary = lower rates + buy securities + reduce reserve requirements. Contractionary = opposite.**

**■ Common Trap: Reserve requirements are rarely changed but candidates must know: lowering them is expansionary, raising is contractionary.**

---

**Q19** Difficulty: Easy LOS: Economics — Elasticity

The price of a good increases from \$10 to \$12, and quantity demanded falls from 100 units to 84 units. The price elasticity of demand is closest to:

A. -0.80
B. -1.00
C. -0.75

**✓ Correct Answer: B — -1.00**

## Explanation:

% change in Qd = (84-100)/100 = -16%. % change in P = (12-10)/10 = 20%. Price elasticity = -16%/20% = -0.80. The closest answer is A (-0.80). Note: If using midpoint method: $\%\Delta Q = -16/92 = -17.4\%$, $\%\Delta P = 2/11 = 18.2\%$, elasticity = -0.956 ≈ -1.00 (B).

**■ Exam Tip: Know both the simple formula and midpoint method. CFA exams often specify which to use.**

**■ Common Trap: The sign of price elasticity is always negative (law of demand). Questions may ask for absolute value.**

---

**Q20** Difficulty: Difficult LOS: Economics — Fiscal Multiplier

In a simple Keynesian model, the marginal propensity to consume (MPC) is 0.75. The government increases spending by \$200 million. What is the total increase in GDP?

A. \$200 million
B. \$600 million
C. \$800 million

**✓ Correct Answer: C — \$800 million**

## Explanation:

Fiscal multiplier = 1 / (1 - MPC) = 1 / (1 - 0.75) = 1 / 0.25 = 4. Total $\Delta$GDPs = Multiplier × $\Delta$G = 4 × \$200M = \$800M. The initial \$200M spending triggers rounds of consumption: recipients spend 75%, those recipients spend 75% of that, and so on.

**■ Exam Tip: Fiscal multiplier = 1/(1-MPC) = 1/MPS. Higher MPC = larger multiplier effect.**

**■ Common Trap: Candidates often answer \$200M (no multiplier) or \$600M (multiplier of 3). The multiplier is 4 with MPC=0.75.**

| |
|---|
| **GDP = C + I + G + (X-M) — Expenditure approach to GDP** |
| **Fiscal Multiplier = 1/(1-MPC) — Government spending multiplier** |
| **PED = %ΔQd / %ΔP — Price elasticity of demand** |
| **Money Multiplier = 1/Reserve Requirement — Banking system money creation** |

CHAPTER 4 Exam Weight: 13–17%

# Financial Reporting & Analysis

FRA is the second highest-weighted topic on Level 1. It covers the three financial statements, financial ratios, revenue recognition, and key accounting differences between IFRS and US GAAP. A systematic understanding of how transactions flow through the statements is essential.

**Q21** Difficulty: Easy LOS: Financial Reporting & Analysis — Balance Sheet

A company reports total assets of $500,000 and total liabilities of $320,000. Shareholders' equity is:

A. $820,000
B. $180,000
C. $320,000

**✓ Correct Answer: B — $180,000**

## Explanation:

Accounting equation: Assets = Liabilities + Equity. Therefore Equity = Assets - Liabilities = 500,000 - 320,000 = $180,000. This is the basic balance sheet equation and forms the foundation of financial analysis.

**■ Exam Tip: Assets = Liabilities + Equity. Always start here. Equity is the residual claim on assets.**

**■ Common Trap: Option A ($820,000) is Assets + Liabilities — a common calculation error. Never add them.**

**Q22** Difficulty: Medium LOS: Financial Reporting & Analysis — Income Statement

A company has revenues of $1,000,000, COGS of $600,000, operating expenses of $150,000, interest expense of $30,000, and a tax rate of 30%. Net income is closest to:

A. $154,000
B. $220,000
C. $175,000

**✓ Correct Answer: A — $154,000**

## Explanation:

Gross profit = 1,000,000 - 600,000 = $400,000. EBIT = 400,000 - 150,000 = $250,000. EBT = 250,000 - 30,000 = $220,000. Net income = 220,000 × (1 - 0.30) = $154,000. Option B ($220,000) is pre-tax income — a common error.

**■ Exam Tip: Work top-down: Revenue → Gross Profit → EBIT → EBT → Net Income.**

**■ Common Trap: Candidates who select $220,000 forgot to apply the tax rate. Always apply taxes last.**

**Q23** Difficulty: Medium LOS: Financial Reporting & Analysis — Cash Flow Statement

Under IFRS, which of the following items gives companies the most flexibility in classification on the cash flow statement?

A. Cash from operating activities.
B. Interest paid and dividends received.
C. Proceeds from issuing common stock.

**✓ Correct Answer: B — Interest paid and dividends received**

## Explanation:

Under IFRS, interest paid can be classified as operating or financing, and dividends received can be classified as operating or investing. Under US GAAP, both are classified as operating. This IFRS flexibility allows companies to manage how their operating cash flows appear.

**■ Exam Tip: IFRS is more flexible than US GAAP for interest/dividend classification. Know the differences for the exam.**

**■ Common Trap: Proceeds from issuing stock (C) is always financing under both IFRS and US GAAP — no flexibility there.**

---

**Q24** Difficulty: Difficult LOS: Financial Reporting & Analysis — Financial Ratios

Company X has current assets of $800,000, inventories of $300,000, prepaid expenses of $50,000, and current liabilities of $400,000. The quick ratio is:

A. 2.00
B. 1.13
C. 1.75

**✓ Correct Answer: B — 1.13**

## Explanation:

Quick Ratio = (Current Assets - Inventories - Prepaid Expenses) / Current Liabilities = (800,000 - 300,000 - 50,000) / 400,000 = 450,000 / 400,000 = 1.125 ≈ 1.13. The quick ratio excludes less liquid assets like inventories and prepaid expenses.

**■ Exam Tip: Quick ratio excludes inventories AND prepaid expenses. Current ratio includes all current assets.**

**■ Common Trap: Candidates often only subtract inventories and forget prepaid expenses. Both must be excluded.**

---

**Q25** Difficulty: Medium LOS: Financial Reporting & Analysis — Revenue Recognition

Under IFRS 15 (Revenue from Contracts with Customers), revenue should be recognized when:

A. Cash is received from the customer.
B. The company satisfies a performance obligation by transferring control of a good or service.
C. A legally binding contract is signed with the customer.

**✓ Correct Answer: B — The company satisfies a performance obligation by transferring control**

## Explanation:

IFRS 15 (and ASC 606 under US GAAP) require revenue recognition when (or as) performance obligations are satisfied — i.e., when control of the promised good or service transfers to the customer. This is a 5-step model: (1) identify the contract, (2) identify performance obligations, (3) determine transaction price, (4) allocate price, (5) recognize when obligation satisfied.

**■ Exam Tip: Revenue recognition = transfer of control, not signing of contract or receipt of cash.**

**■ Common Trap: Signing a contract (C) is step 1 but NOT when revenue is recognized. Cash receipt (A) is the old cash-basis method.**

---

**Q26** Difficulty: Easy LOS: Financial Reporting & Analysis — Depreciation

A machine costs $100,000, has a salvage value of $10,000, and a useful life of 5 years. Using straight-line depreciation, the annual depreciation expense is:

A. $20,000
B. $18,000
C. $22,000

**✓ Correct Answer: B — $18,000**

## Explanation:

Straight-line depreciation = (Cost - Salvage Value) / Useful Life = (100,000 - 10,000) / 5 = 90,000 / 5 = $18,000 per year. Option A ($20,000) ignores the salvage value.

**■ Exam Tip: Straight-line: (Cost - Salvage) / Life. Never forget to subtract salvage value.**

**■ Common Trap: Forgetting salvage value is the classic trap, giving $20,000 instead of $18,000.**

---

**Q27** Difficulty: Difficult LOS: Financial Reporting & Analysis — DuPont Analysis

Using the 3-factor DuPont decomposition, a company has net profit margin of 8%, total asset turnover of 1.5, and equity multiplier of 2.5. Return on equity (ROE) is:

A. 20%
B. 30%
C. 12%

**✓ Correct Answer: B — 30%**

## Explanation:

ROE = Net Profit Margin × Total Asset Turnover × Equity Multiplier = 0.08 × 1.5 × 2.5 = 0.30 = 30%. The DuPont model shows that ROE is driven by profitability (margin), efficiency (turnover), and leverage (equity multiplier).

**■ Exam Tip: ROE (DuPont) = Profit Margin × Asset Turnover × Equity Multiplier. This is the 3-factor model.**

**■ Common Trap: Candidates sometimes only multiply two of the three factors. All three must be included.**

---

| |
|---|
| **Assets = Liabilities + Equity — Fundamental balance sheet equation** |
| **Gross Profit = Revenue - COGS — Step 1 of income statement** |
| **Net Income = EBT × (1 - Tax Rate) — Bottom line after tax** |
| **Quick Ratio = (CA - Inv - Prepaid) / CL — Liquidity excluding illiquid current assets** |
| **SL Depreciation = (Cost - Salvage) / Life — Annual straight-line depreciation** |
| **ROE = Margin × Turnover × Equity Multiplier — DuPont 3-factor decomposition** |

CHAPTER 5 Exam Weight: 10–12%

# Equity Investments

Equity Investments covers equity valuation models, market efficiency, industry analysis, and equity indices. Key valuation methods include the DDM, EV/EBITDA, and P/E ratios. Understanding market efficiency is also essential for interpreting why anomalies exist.

**Q28** Difficulty: Easy LOS: Equity Investments — P/E Ratio

A company earns $4.00 per share and trades at $60 per share. The price-to-earnings (P/E) ratio is:

A. 15x
B. 20x
C. 10x

**✓ Correct Answer: A — 15x**

### Explanation:

P/E = Price / EPS = $60 / $4 = 15x. This means investors are paying $15 for every $1 of earnings. A high P/E may indicate growth expectations; a low P/E may indicate undervaluation or low growth prospects.

**■ Exam Tip: P/E = Price per share / EPS. The "x" denotes times earnings.**

**■ Common Trap: Candidates sometimes invert the ratio, calculating EPS/Price (earnings yield). P/E = Price OVER EPS.**

**Q29** Difficulty: Medium LOS: Equity Investments — DDM Valuation

A stock pays an annual dividend of $3.00. Dividends are expected to grow at 5% per year indefinitely. The required return is 12%. The intrinsic value using the Gordon Growth Model is closest to:

A. $42.86
B. $37.50
C. $25.00

**✓ Correct Answer: A — $42.86**

### Explanation:

Gordon Growth Model: P0 = D1 / (r - g) where D1 = D0 × (1+g) = 3.00 × 1.05 = $3.15. P0 = 3.15 / (0.12 - 0.05) = 3.15 / 0.07 = $45.00. Wait — let me recalculate: if D0 = $3.00, D1 = $3.15, P = 3.15/0.07 = $45. If D1 = $3.00 (already next year's dividend), P = 3.00/0.07 = $42.86. The question says "annual dividend of $3.00" which is D1, so P = 3.00/(0.12-0.05) = $42.86.

**■ Exam Tip: GGM: P0 = D1/(r-g). If given current dividend D0, must grow it: D1 = D0(1+g).**

**■ Common Trap: The most common error: using D0 instead of D1. Always check whether the dividend given is current or next period.**

**Q30** Difficulty: Medium LOS: Equity Investments — EV/EBITDA

A firm has a market cap of $500M, debt of $200M, cash of $50M, and EBITDA of $100M. Its EV/EBITDA multiple is:

A. 5.0x

B. 6.5x
C. 5.5x

**✓ Correct Answer: B — 6.5x**

## Explanation:

Enterprise Value (EV) = Market Cap + Debt - Cash = 500 + 200 - 50 = $650M. EV/EBITDA = 650/100 = 6.5x. EV/EBITDA is useful for comparing companies with different capital structures because it is capital structure neutral.

**■ Exam Tip: EV = Market Cap + Debt - Cash (minus cash because an acquirer gets it). Then divide by EBITDA.**

**■ Common Trap: Forgetting to subtract cash is the most common EV calculation error. Cash reduces the effective price of acquisition.**

---

**Q31** Difficulty: Difficult LOS: Equity Investments — Market Efficiency

An analyst discovers that stocks with low price-to-book ratios consistently outperform the market over 10-year periods, even after adjusting for risk. This evidence is most inconsistent with:

A. Weak-form market efficiency.
B. Semi-strong-form market efficiency.
C. Strong-form market efficiency.

**✓ Correct Answer: B — Semi-strong-form market efficiency**

## Explanation:

P/B ratios are publicly available information. If a strategy based on publicly available information (like P/B ratios) consistently generates abnormal returns, this violates semi-strong form efficiency, which states that all public information is already reflected in prices. Weak form only covers historical price data. Strong form covers all information including private.

**■ Exam Tip: Semi-strong = public info. If public info generates excess returns → semi-strong is violated.**

**■ Common Trap: Candidates often select strong-form. But P/B is public info, not private — so it's semi-strong that's violated.**

---

| |
|---|
| **P/E = Price / EPS — Price-to-earnings ratio** |
| **P0 = D1 / (r - g) — Gordon Growth Model (GGM)** |
| **EV = Mkt Cap + Debt - Cash — Enterprise value** |
| **EV/EBITDA = EV / EBITDA — Capital structure-neutral valuation multiple** |

CHAPTER 6 Exam Weight: 10–12%

# Fixed Income

Fixed Income covers bond pricing, yield measures, duration, convexity, and the term structure of interest rates. The inverse relationship between bond prices and yields is the cornerstone concept. Duration measures interest rate risk; convexity improves the approximation.

**Q32** Difficulty: Easy LOS: Fixed Income — Bond Pricing

A 5-year bond with a face value of $1,000 and a coupon rate of 6% (paid annually) is priced to yield 8%. The price of the bond is closest to:

A. $920.15
B. $1,000.00
C. $1,080.00

**✓ Correct Answer: A — $920.15**

## Explanation:

When the coupon rate (6%) < yield (8%), the bond trades at a discount (below par). Bond Price = Sum of PV of coupons + PV of face. PV of coupons = 60 × [1-(1.08)^-5]/0.08 = 60 × 3.9927 = $239.56. PV of face = 1000/(1.08)^5 = 1000/1.4693 = $680.58. Price = 239.56 + 680.58 = $920.14.

**■ Exam Tip: Coupon > Yield → premium. Coupon < Yield → discount. Coupon = Yield → par. Memorize this rule.**

**■ Common Trap: Many candidates select par ($1,000) when they see "6% coupon." The yield determines the price, not the coupon alone.**

**Q33** Difficulty: Medium LOS: Fixed Income — Duration

A bond has a modified duration of 5.5 and a yield-to-maturity of 7%. If yields increase by 50 basis points, the approximate percentage change in bond price is:

A. +2.75%
B. -2.75%
C. -3.50%

**✓ Correct Answer: B — -2.75%**

## Explanation:

%ΔPrice ≈ -Modified Duration × ΔYield = -5.5 × 0.0050 = -0.0275 = -2.75%. Yield increases → price decreases (inverse relationship). The modified duration formula gives the approximate, linear price change.

**■ Exam Tip: %ΔP ≈ -Modified Duration × ΔYield. Yield up → price down → negative sign.**

**■ Common Trap: Candidates forget the negative sign. Bond prices move opposite to yields.**

**Q34** Difficulty: Medium LOS: Fixed Income — Yield Curve

The yield curve is upward sloping (normal). The most likely explanation according to the liquidity preference theory is:

A. Markets expect interest rates to fall in the future.

B. Investors require a liquidity premium for holding longer-term bonds.
C. Supply and demand for funds at each maturity is equal.

**✓ Correct Answer: B — Investors require a liquidity premium for holding longer-term bonds**

## Explanation:

The liquidity preference theory states that investors prefer short-term (more liquid) securities and must be compensated with a liquidity premium to hold longer-term bonds. This premium increases with maturity, resulting in an upward-sloping yield curve even if rates are expected to stay flat.

**■ Exam Tip: Liquidity preference theory → upward slope = liquidity premium, not necessarily expectations of rate increases.**

**■ Common Trap: Candidates confuse pure expectations theory (shape = only rate expectations) with liquidity preference (shape = expectations + liquidity premium).**

---

| **Q35** | Difficulty: Difficult | LOS: Fixed Income — Convexity |
|---|---|---|

Two bonds have the same modified duration of 6. Bond A has positive convexity; Bond B has negative convexity (callable bond). When yields fall by 100 bps, which bond gains more in price?

A. Bond B gains more because callable bonds have lower duration.
B. Bond A gains more because positive convexity benefits the bondholder when yields fall.
C. Both bonds gain the same amount because they have the same duration.

**✓ Correct Answer: B — Bond A gains more because positive convexity benefits the bondholder**

## Explanation:

With equal duration, convexity determines the second-order price effect. Positive convexity means the price gain from falling yields is greater than the price loss from rising yields by the same amount. Negative convexity (callable bonds) caps the price gain because the issuer will call the bond when rates fall. Bond A benefits more from the yield decline.

**■ Exam Tip: Positive convexity: gains more when yields fall, loses less when yields rise. Always beneficial.**

**■ Common Trap: Candidates think "same duration = same price change." Duration is the first-order approximation; convexity is the critical second-order difference.**

---

| |
|---|
| **Bond Price = Σ PV(coupons) + PV(face) — Present value of all cash flows** |
| **%ΔPrice ≈ -Mod.Duration × ΔYield — Duration-based price change approximation** |
| **Coupon > Yield → Premium \| Coupon < Yield → Discount — Bond pricing rule** |
| **P = C + PV(X) for Put-Call Parity — Options parity: Call + PV(Strike) = Put + Stock** |

CHAPTER 7 Exam Weight: 5–8%

# Derivatives

Derivatives includes forwards, futures, options, and swaps. Key concepts include put-call parity, option payoff diagrams, and the differences between exchange-traded and OTC instruments.

**Q36** Difficulty: Easy LOS: Derivatives — Call Option

An investor buys a call option with a strike price of $50 for a premium of $3. At expiration, the stock price is $58. The profit per share is:

A. $8
B. $5
C. $11

**✓ Correct Answer: B — $5**

## Explanation:

Call option payoff = Max(Stock Price - Strike, 0) = Max(58 - 50, 0) = $8. Profit = Payoff - Premium = 8 - 3 = $5 per share. The option is in-the-money by $8, but the initial premium of $3 reduces the profit.

**■ Exam Tip: Call profit = Max(S-X, 0) - Premium. Always subtract the premium paid.**

**■ Common Trap: Candidates answer $8 (the payoff) without subtracting the premium. Profit ≠ Payoff.**

**Q37** Difficulty: Medium LOS: Derivatives — Put-Call Parity

A stock trades at $100. A European call with strike $100 expiring in 1 year costs $10. The risk-free rate is 5%. Using put-call parity, the price of a European put with the same terms is closest to:

A. $5.24
B. $10.00
C. $4.76

**✓ Correct Answer: A — $5.24**

## Explanation:

Put-Call Parity: C + PV(X) = P + S. PV(X) = 100/1.05 = $95.24. So: 10 + 95.24 = P + 100. P = 105.24 - 100 = $5.24.

**■ Exam Tip: Put-Call Parity: C + PV(X) = P + S. Rearrange to find any missing variable.**

**■ Common Trap: Candidates often use X directly ($100) instead of PV(X) ($95.24). Always discount the strike price.**

**Q38** Difficulty: Difficult LOS: Derivatives — Futures vs. Forwards

Compared to forward contracts, futures contracts are least likely to have:

A. Standardized contract terms.
B. Daily mark-to-market settlement.
C. Counterparty credit risk.

**✓ Correct Answer: C — Counterparty credit risk**

### Explanation:

Futures are exchange-traded and cleared through a central clearinghouse, which virtually eliminates counterparty credit risk through margin requirements and daily settlement. Forwards are OTC contracts with significant counterparty risk. Futures ARE standardized (A) and DO have daily mark-to-market (B) — so these are NOT things futures "least likely" have.

■ **Exam Tip: Futures: exchange-traded, standardized, mark-to-market daily, minimal credit risk. Forwards: OTC, customized, credit risk.**

■ **Common Trap: The question asks what futures are LEAST LIKELY to have. Counterparty credit risk is eliminated in futures — so that's the answer.**

---

| |
|---|
| **Call Payoff = Max(S-X, 0) — Long call payoff at expiration** |
| **Put Payoff = Max(X-S, 0) — Long put payoff at expiration** |
| **C + PV(X) = P + S — Put-call parity** |
| **Profit = Payoff - Premium Paid — Net profit on long options** |

CHAPTER 8 Exam Weight: 5–8%

# Portfolio Management

Portfolio Management covers modern portfolio theory, CAPM, the efficient frontier, and the Capital Market Line. Key metrics include Sharpe ratio, Treynor ratio, and Jensen's alpha.

**Q39** Difficulty: Easy LOS: Portfolio Management — Risk and Return

An investor holds two assets. Asset A has an expected return of 10% and weight 60%. Asset B has an expected return of 15% and weight 40%. The expected portfolio return is:

A. 12.0%
B. 12.5%
C. 13.0%

**✓ Correct Answer: A — 12.0%**

## Explanation:

E(Rp) = w_A × E(R_A) + w_B × E(R_B) = 0.60 × 10% + 0.40 × 15% = 6% + 6% = 12%.

**■ Exam Tip: Portfolio expected return is the weighted average of individual asset returns.**

**■ Common Trap: Candidates sometimes simply average (10+15)/2 = 12.5%. Always weight by portfolio allocation.**

**Q40** Difficulty: Medium LOS: Portfolio Management — Capital Market Line

The Capital Market Line (CML) is best described as the set of:

A. All possible portfolios that can be formed from risky assets.
B. Portfolios formed by combining the risk-free asset and the market portfolio.
C. Portfolios with the minimum variance for each level of expected return.

**✓ Correct Answer: B — Portfolios formed by combining the risk-free asset and the market portfolio**

## Explanation:

The CML represents all efficient portfolios formed by blending the risk-free asset and the tangency (market) portfolio. Moving along the CML by changing the allocation between risk-free and market portfolio gives different risk-return combinations. Option A describes the efficient frontier of risky assets. Option C describes the minimum-variance frontier.

**■ Exam Tip: CML = risk-free asset + market portfolio. It is the efficient frontier when a risk-free asset exists.**

**■ Common Trap: Candidates confuse CML (includes risk-free) with efficient frontier (risky assets only). The CML dominates the efficient frontier.**

**Q41** Difficulty: Difficult LOS: Portfolio Management — Beta

A portfolio has a beta of 1.3 relative to the market. The market is expected to return 10% and the risk-free rate is 3%. Using CAPM, the required return on the portfolio is:

A. 12.1%
B. 9.1%
C. 13.0%

✓ **Correct Answer: A — 12.1%**

## Explanation:

CAPM: $E(R_i) = R_f + \beta_i \times (E(R_m) - R_f) = 3\% + 1.3 \times (10\% - 3\%) = 3\% + 1.3 \times 7\% = 3\% + 9.1\% = 12.1\%$.

■ **Exam Tip: CAPM: Required Return = Rf + Beta × (Market Return - Rf). The (Rm - Rf) is the equity risk premium.**

■ **Common Trap: Candidates sometimes compute Beta × Rm instead of Beta × (Rm - Rf). Always use the equity risk premium, not the market return directly.**

| |
|---|
| **$E(R_p) = \Sigma w_i \times E(R_i)$ — Portfolio expected return (weighted average)** |
| **CAPM: $E(R_i) = R_f + \beta(R_m - R_f)$ — Required return using Capital Asset Pricing Model** |
| **Sharpe = $(R_p - R_f) / \sigma_p$ — Reward-to-total-risk ratio** |
| **Treynor = $(R_p - R_f) / \beta_p$ — Reward-to-systematic-risk ratio** |

CHAPTER 9 Exam Weight: 5–8%

# Alternative Investments

Alternative Investments covers hedge funds, private equity, real estate, commodities, and infrastructure. Focus on fee structures, valuation methods, and risk-return characteristics.

**Q42** Difficulty: Easy LOS: Alternative Investments — Hedge Fund Fees

A hedge fund charges "2 and 20" fees. An investor invests $1,000,000 and the fund generates a 30% gross return. Assuming a high water mark equals the initial investment, the management fee is $20,000 and the incentive fee is:

A. $60,000
B. $56,000
C. $54,000

**✓ Correct Answer: B — $56,000**

## Explanation:

Gross return = $300,000. Management fee = 2% × $1,000,000 = $20,000. After management fee, net gain = $300,000 - $20,000 = $280,000. Incentive fee = 20% × $280,000 = $56,000. Total fees = $76,000.

**■ Exam Tip: "2 and 20": 2% management fee on NAV, 20% performance fee on profits (after management fee in many structures).**

**■ Common Trap: Some candidates apply the 20% to the full gross return ($300,000 × 20% = $60,000). Always check whether management fees reduce the base for incentive calculations.**

**Q43** Difficulty: Medium LOS: Alternative Investments — Real Estate

A property generates net operating income (NOI) of $150,000 per year. Comparable properties trade at a cap rate of 6%. The estimated value using direct capitalization is:

A. $9,000
B. $2,500,000
C. $900,000

**✓ Correct Answer: B — $2,500,000**

## Explanation:

Value = NOI / Cap Rate = $150,000 / 0.06 = $2,500,000. The capitalization rate is the required rate of return on real estate. Lower cap rates indicate higher valuations (less risk), similar to lower P/E for stocks.

**■ Exam Tip: Real Estate Value = NOI / Cap Rate. Lower cap rate = higher value. Higher cap rate = lower value.**

**■ Common Trap: Candidates sometimes multiply NOI × Cap Rate: 150,000 × 0.06 = $9,000 — this is wrong. Divide NOI by cap rate.**

**RE Value = NOI / Cap Rate — Real estate direct capitalization**

| |
|---|
| **Hedge Fund Fees: 2% mgmt + 20% performance — "2 and 20" structure** |
| **NAV = Assets - Liabilities — Net asset value of fund** |

CHAPTER 10 Exam Weight: 2–5%

# Behavioral Finance

Behavioral Finance examines how psychological biases affect investor decisions and market prices. Key biases include loss aversion, overconfidence, herding, and the disposition effect.

**Q44** **Difficulty: Easy** LOS: Behavioral Finance — Cognitive Biases

An investor refuses to sell a stock that has declined 40% because she believes it will recover to her purchase price before she will accept a loss. This behavior is best described as:

A. Overconfidence bias.
B. Loss aversion / Disposition effect.
C. Anchoring bias.

**✓ Correct Answer: B — Loss aversion / Disposition effect**

## Explanation:

The disposition effect describes the tendency to hold losing investments too long (hoping to break even) and sell winners too soon (locking in gains). This stems from loss aversion — the pain of realizing a loss exceeds the pleasure of an equivalent gain. The investor is anchoring on the purchase price as a reference point.

**■ Exam Tip: Disposition effect = hold losers too long, sell winners too soon. Driven by loss aversion.**

**■ Common Trap: Anchoring (C) is also present here, but the primary bias being described is the disposition effect / loss aversion. The question asks for the best description.**

**Q45** **Difficulty: Medium** LOS: Behavioral Finance — Market Anomalies

Research consistently shows that small-cap stocks earn higher risk-adjusted returns than large-cap stocks over long periods. This is most consistent with:

A. The efficient market hypothesis.
B. A risk-based explanation — small-cap stocks have higher systematic risk.
C. Behavioral bias — investors underestimate small company growth.

**✓ Correct Answer: B — A risk-based explanation**

## Explanation:

The size effect (small-cap premium) is documented in financial literature. The debate is whether it reflects: (1) higher risk (the Fama-French model treats it as a risk factor) or (2) mispricing/behavioral bias. The CFA curriculum presents the size premium as a potential risk factor, though both explanations have empirical support.

**■ Exam Tip: The size premium can be explained by either higher risk or behavioral mispricing. Know both arguments.**

**■ Common Trap: This is a nuanced question. The EMH (A) would say the premium reflects risk — which overlaps with (B). The best answer is B for this context.**

**Disposition Effect — Hold losers too long; sell winners too soon**

| |
|---|
| **Loss Aversion — Pain of loss > pleasure of equal gain** |
| **Overconfidence — Overestimate own ability; underestimate risk** |
| **Anchoring — Rely too heavily on first piece of information** |

CHAPTER 11 Exam Weight: 2–5%

# GIPS — Global Investment Performance Standards

GIPS standards ensure fair representation and full disclosure of investment performance. Key concepts: composite construction, minimum 5-year history, verification.

**Q46** Difficulty: Medium LOS: GIPS — Composite Construction

Under GIPS, which of the following statements about composite construction is most accurate?

A. Firms may exclude non-discretionary accounts from composites at their discretion.
B. All fee-paying, discretionary accounts must be included in at least one composite.
C. Composites should include only the firm's largest and most representative accounts.

**✓ Correct Answer: B — All fee-paying, discretionary accounts must be included in at least one composite**

## Explanation:

GIPS requires that all actual, discretionary, fee-paying accounts be assigned to at least one composite. This prevents cherry-picking of best-performing accounts. Non-discretionary accounts are excluded because the firm doesn't control investment decisions, but all qualifying accounts must be in a composite.

**■ Exam Tip: GIPS composite rule: ALL discretionary, fee-paying accounts → at least one composite. No exceptions.**

**■ Common Trap: Candidates think firms can choose which accounts to include. Under GIPS, all qualifying accounts must be included — no cherry-picking.**

**Q47** Difficulty: Easy LOS: GIPS — Presentation Requirements

Under GIPS, a firm must present at least how many years of GIPS-compliant performance history, or the period since inception if shorter?

A. 3 years
B. 5 years
C. 10 years

**✓ Correct Answer: B — 5 years**

## Explanation:

GIPS requires firms to present a minimum of 5 years of compliant performance (or since inception if the firm/composite has less than 5 years of history). After the initial 5 years, firms must add one additional year each year until they reach 10 years, then maintain 10 years of history.

**■ Exam Tip: GIPS minimum: 5 years initially, build up to 10 years. After 10 years: always show 10.**

**■ Common Trap: Many candidates say 3 or 10. Initial minimum is 5 years, with a build-up to 10. Both numbers matter.**

| |
|---|
| **Composite — Aggregation of discretionary fee-paying accounts with similar strategy** |
| **Minimum History — 5 years initially; build to 10 years** |
| **Verification — Third-party review of compliance — voluntary but recommended** |

CHAPTER 12 Exam Weight: 8–10%

# Corporate Issuers

Corporate Issuers covers capital structure theory, dividend policy, working capital, and the cost of capital. MM theorems form the theoretical foundation; WACC is the key metric.

**Q48** Difficulty: Easy LOS: Corporate Issuers — Capital Structure

The Modigliani-Miller theorem without taxes states that:

A. A firm's value increases as it adds more debt due to tax shields.
B. A firm's value is independent of its capital structure.
C. Equity financing is always preferable to debt financing.

**✓ Correct Answer: B — A firm's value is independent of its capital structure**

## Explanation:

MM Proposition I (no taxes): In a perfect market (no taxes, transaction costs, or financial distress), the total value of a firm is independent of how it is financed. Whether the firm uses debt or equity does not affect firm value. This is the foundation — the real world then adds taxes (tax shields) and financial distress costs.

**■ Exam Tip: MM (no taxes): Capital structure irrelevant. MM (with taxes): Debt adds value via tax shields.**

**■ Common Trap: MM WITH taxes (A) is different from MM WITHOUT taxes (B). The question specifies "without taxes" — so value is independent of capital structure.**

**Q49** Difficulty: Medium LOS: Corporate Issuers — Dividend Policy

A company follows a residual dividend policy. It has $5 million in capital spending needs, a target 40% debt ratio, and net income of $4 million. The dividend payment is closest to:

A. $1,000,000
B. $2,000,000
C. $0

**✓ Correct Answer: A — $1,000,000**

## Explanation:

Equity needed for capex = Capital budget × (1 - Debt ratio) = $5M × 0.60 = $3M. Residual dividend = Net income - Equity needed = $4M - $3M = $1,000,000. Under residual dividend policy, dividends are paid only from earnings left after funding all positive-NPV investments.

**■ Exam Tip: Residual dividend = Net Income - (Equity portion of capital budget). Equity portion = Capex × (1 - D/A).**

**■ Common Trap: Candidates forget to apply the equity ratio to the capex. The firm uses debt to fund 40% of capex, so only 60% comes from equity (retained earnings).**

**Q50** Difficulty: Medium LOS: Corporate Issuers — Weighted Average Cost of Capital

A firm has 40% equity (cost 12%) and 60% debt (pre-tax cost 8%, tax rate 30%). The WACC is:

A. 6.72%
B. 8.16%

C. 9.60%

✓ **Correct Answer: B — 8.16%**

## Explanation:

WACC = w_e × r_e + w_d × r_d × (1 - tax) = 0.40 × 12% + 0.60 × 8% × (1-0.30) = 4.8% + 0.60 × 5.6% = 4.8% + 3.36% = 8.16%.

■ **Exam Tip: WACC: weight equity by cost of equity, weight debt by after-tax cost of debt. Always use after-tax cost of debt.**

■ **Common Trap: Using pre-tax cost of debt (8% instead of 5.6%) gives 4.8% + 4.8% = 9.6% — a very common mistake.**

---

| |
|---|
| **WACC = we×re + wd×rd×(1-t) — Weighted average cost of capital** |
| **MM (no tax): Value independent of capital structure — Modigliani-Miller Prop I** |
| **Residual Div = NI - (Equity × Capex) — Residual dividend policy** |
| **After-tax cost of debt = rd × (1 - tax rate) — Tax shield reduces cost of debt** |

MOCK EXAM

# 15-Question Practice Exam

Instructions: Allow 27 minutes (1.5 minutes per question). Answer all questions before checking the answer key on the following page. This exam covers all 12 topics proportionally.

---

### Question 1

The CFA Standards require a member to:

A. Always comply with the CFA Code regardless of local law.
B. Comply with the stricter of local law and the CFA Standards.
C. Report all violations to the CFA Institute within 30 days.

### Question 2

A portfolio has E(R) = 12% and standard deviation of 15%. The Sharpe ratio with Rf = 3% is:

A. 0.60
B. 0.80
C. 0.75

### Question 3

Under IFRS, development costs are:

A. Expensed as incurred.
B. Capitalized when criteria are met.
C. Amortized over 5 years.

### Question 4

Bond A has duration 4 and Bond B has duration 7. When yields rise by 100 bps, which bond loses more value?

A. Bond A
B. Bond B
C. They lose equally

### Question 5

A firm's current ratio is 2.5 and quick ratio is 1.2. This suggests:

A. The firm has high inventory relative to other current assets.
B. The firm has no inventory.
C. The firm is illiquid.

### Question 6

Which of the following is a money market security?

A. 10-year Treasury bond
B. 90-day T-Bill
C. Corporate bond due in 18 months

### Question 7

The price of a stock using the Gordon Growth Model increases when:

A. The discount rate increases.
B. The dividend growth rate increases.
C. The dividend payout decreases.

### Question 8

A futures contract differs from a forward contract in that futures:

A. Have customizable terms.
B. Are marked to market daily.
C. Have higher counterparty risk.

### Question 9

GDP measured from the expenditure approach is:

A. C + I + G + (X-M)
B. C + I + G - T
C. C + S + T

### Question 10

Under GIPS, a composite must include:

A. All discretionary fee-paying accounts.
B. Only the top quartile accounts.
C. Accounts over $1 million only.

### Question 11

An ETF is best described as:

A. A closed-end fund that issues fixed shares at IPO.
B. An index fund that trades on an exchange like a stock.
C. An actively managed mutual fund.

### Question 12

The equity risk premium is:

A. Expected market return minus the risk-free rate.
B. Expected market return plus the risk-free rate.
C. Beta times the expected market return.

### Question 13

Straight-line depreciation of a $50,000 asset with $5,000 salvage value over 5 years is:

A. $10,000/year
B. $9,000/year
C. $11,000/year

### Question 14

An investor buys a put option with X=$40 for $2. At expiry, S=$35. Profit per share is:

A. $3
B. $5
C. -$2

### Question 15

The efficient market hypothesis in semi-strong form states that prices reflect:

A. All historical price data only.
B. All publicly available information.
C. All public and private information.

## MOCK EXAM — ANSWER KEY

Check your answers below. For explanations, refer to the relevant chapter.

| Q# | Answer | Topic |
|---|---|---|
| 1 | **B** | **Ethics** |
| 2 | **A** | **Quant** |
| 3 | **B** | **Economics** |
| 4 | **B** | **FRA** |
| 5 | **A** | **Equity** |
| 6 | **B** | **FI** |
| 7 | **B** | **Derivatives** |
| 8 | **B** | **Portfolio** |
| 9 | **A** | **Alternatives** |
| 10 | **A** | **Behavioral** |
| 11 | **B** | **GIPS** |
| 12 | **A** | **Corp** |
| 13 | **B** | **Ethics** |
| 14 | **A** | **Quant** |
| 15 | **B** | **Equity** |

# FORMULA MASTER REFERENCE

This master reference consolidates every key formula tested on CFA Level 1. Review this section before your exam.

## TIME VALUE OF MONEY

**$FV = PV(1+r)^n$ → Future value**

**$PV = FV/(1+r)^n$ → Present value**

**$HPR = (P_1 - P_0 + D_1)/P_0$ → Holding period return**

**$EAR = (1+r/m)^m - 1$ → Effective annual rate**

## STATISTICS & PROBABILITY

**$E(R) = \Sigma\, p_i R_i$ → Expected return**

**$\sigma^2 = \Sigma\, p_i(R_i - E(R))^2$ → Variance**

**$CV = \sigma / E(R)$ → Coefficient of variation**

**$P(A \cap B) = P(A) \times P(B|A)$ → Joint probability**

## PORTFOLIO THEORY

**$E(R_p) = \Sigma\, w_i E(R_i)$ → Portfolio expected return**

**CAPM: $E(R_i) = R_f + \beta(R_m - R_f)$ → Required return**

**Sharpe = $(R_p - R_f)/\sigma_p$ → Sharpe ratio**

**Treynor = $(R_p - R_f)/\beta$ → Treynor ratio**

## FINANCIAL ANALYSIS

**Current Ratio = CA/CL → Liquidity**

**Quick Ratio = (CA-Inv-Prepaid)/CL → Acid test**

**D/E = Total Debt/Total Equity → Leverage**

**ROE = NI/Equity = Margin×Turnover×Multiplier → DuPont**

## EQUITY VALUATION

**$P_0 = D_1/(r-g)$ → Gordon Growth Model**

**P/E = Price/EPS → P/E multiple**

**EV = MktCap + Debt - Cash → Enterprise value**

**FCFE = NI + D&A; - CapEx - ΔNWC + ΔDebt → Free cash flow to equity**

## FIXED INCOME

**$\%\Delta P \approx -ModDur \times \Delta Yield$ → Price change approximation**

**$ModDur = MacDur / (1 + y/m)$ → Modified duration**

**$C + PV(X) = P + S$ → Put-call parity**

**YTM: Solve PV of all cash flows = Price → Yield to maturity**

## CORPORATE FINANCE

**$WACC = w_e \times r_e + w_d \times r_d \times (1-t)$ → Weighted cost of capital**

**After-tax cost of debt = $r_d(1-t)$ → Debt tax shield**

**NPV = $\Sigma$ CF/(1+r)^t - Initial Outlay → Net present value**

**Residual Div = NI - Equity Retained for Capex → Dividend policy**

## Good Luck on Your CFA Level 1 Exam!

This question bank was designed to help you think like the CFA Institute — identifying the most appropriate action, the best explanation, and the correct application of standards and formulas. Focus on understanding the WHY behind every answer, not just memorizing correct choices.

### Study Strategy Reminder:

1. Always read ALL answer choices before selecting — CFA exams are designed to trap hasty readers.
2. For ethics: the most conservative, protective action is usually correct.
3. For calculations: set up the formula before plugging in numbers.
4. For fixed income: remember — yield up = price down.
5. Use practice exams under timed conditions to build exam stamina.